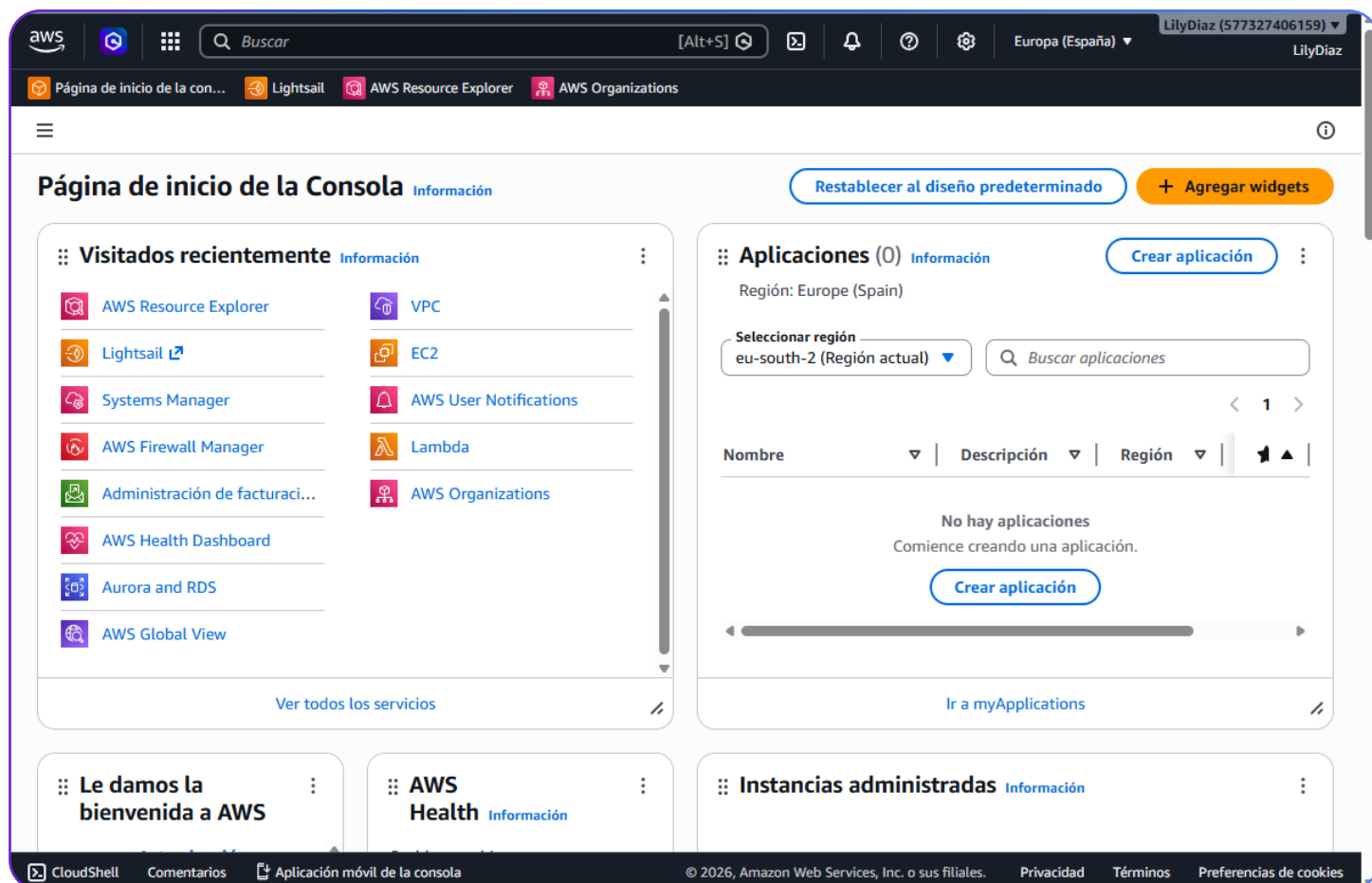


ANEXOS

ANEXO I. Arquitectura técnica del sistema

Amazon Web Services



Despliegue en entorno cloud, infraestructura virtualizada en Amazon Lightsail

The screenshot shows the Amazon Lightsail console. On the left is a navigation menu with options: Instances, Containers, Databases, Networking, Storage, Domains & DNS, Snapshots, Exports, Alarm notifications, Documentation, Lightsail for Research, and Get the mobile app. The main area displays a 'Good evening' greeting and a search bar. Below this, there's a 'Sort by' dropdown set to 'Creation date' and a 'Create instance' button. A single instance is listed: 'cds-wordpress', which is a WordPress instance with 2 GB RAM, 2 vCPUs, and 60 GB SSD, running on a General purpose plan. Its status is 'Running' with a green checkmark. The public IP address is 51.102.208.184. Below the IP, the private IP address 2a05:d014:18e2:7d00:24cb:ac06:5524:1f3c and the location 'Frankfurt, Zone A' are shown.

IP estática

The screenshot shows the details of a static IP address named 'cds-ip'. It is a Static IP, Attached, located in Frankfurt, all zones (eu-central-1). The public static IP address is 51.102.208.184. Below this, there are tabs for 'Details', 'Domains', 'Help & assistance', and 'Delete'. The 'Details' tab is selected, showing the public static IP address and a note that this static IP is available for public connection worldwide. Below this, there's a section titled 'Attach to an instance' with a note that attaching a static IP replaces that instance's dynamic IP address. At the bottom, there's a list of instances with a 'cds-wordpress' instance highlighted, showing its details (2 GB RAM, 2 vCPUs, 60 GB SSD, WordPress) and a 'Detach' button with a red 'X' icon.

Tecnologías utilizadas

WordPress



El proyecto utiliza el CMS WordPress como plataforma principal de gestión de contenidos y desarrollo del entorno web.

WordPress	
Versión	6.9.4
Idioma del sitio	es_ES
Idioma del usuario	es_ES
Zona horaria	Europe/Madrid
URL de inicio	https://cds-advance.es
URL del sitio	https://cds-advance.es
Estructura de enlaces permanentes	/%postname%/
¿Está utilizando HTTPS este sitio?	Sí
¿Es esto un multisitio?	No
¿Cualquiera puede registrarse en este sitio?	No
¿Está el sitio disuadiendo a los motores de búsqueda?	No
Estado por defecto de los comentarios	Abiertos
Tipo de entorno	production
Cantidad de usuarios	13
Comunicación con WordPress.org	Es posible conectarse a WordPress.org

Servidor web: Apache

Lenguaje del servidor: PHP

Servidor

Las opciones que se muestran a continuación se refieren a la configuración de tu servidor. Si hay cambios obligatorios, es posible que necesites la ayuda de tu alojamiento web.

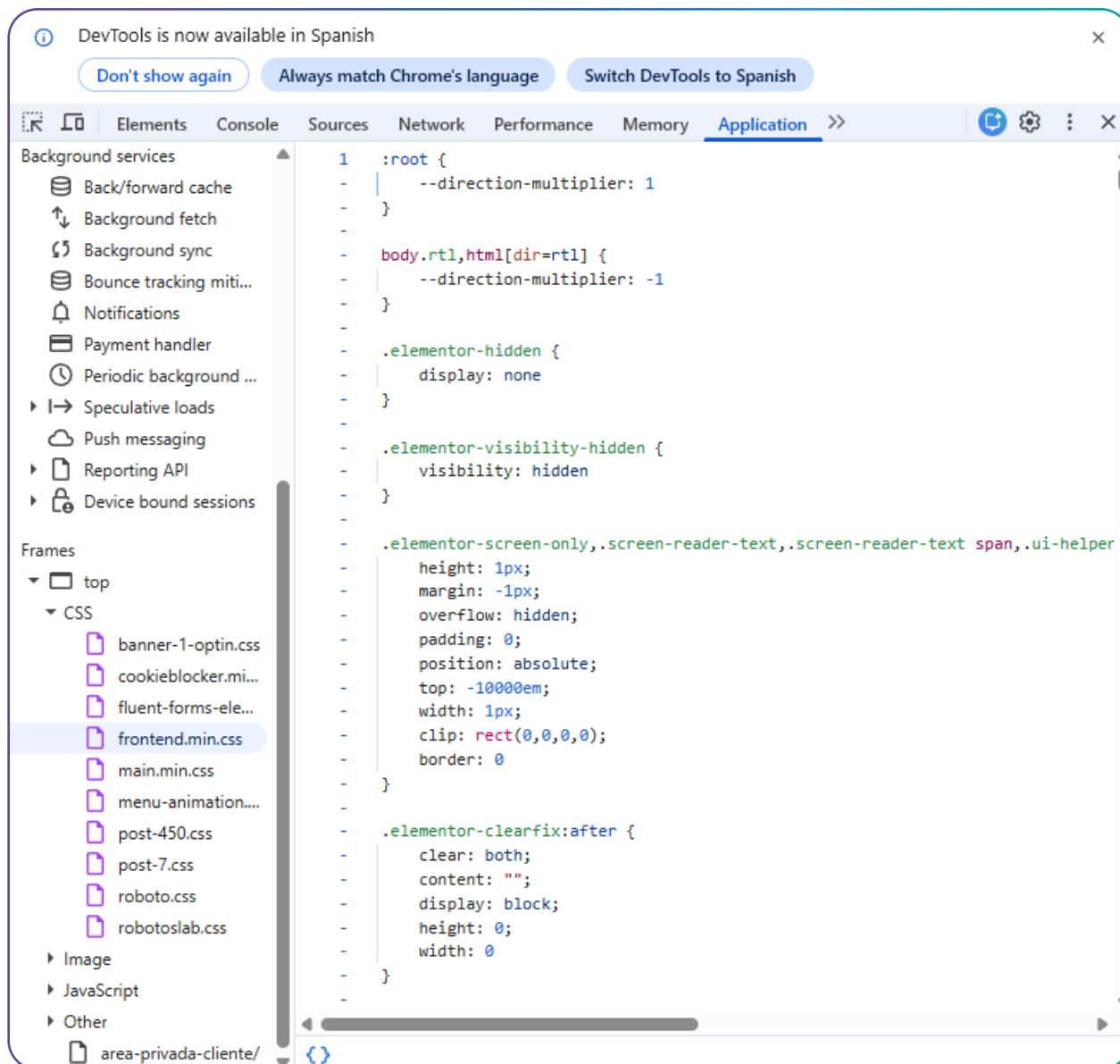
Arquitectura del servidor	Linux 6.1.0-45-cloud-amd64 x86_64
Servidor web	Apache/2.4.67 (Debian)
Versión de PHP	8.2.31 (Admite valores de 64 bits)
PHP SAPI	apache2handler
PHP max input variables	1000
Límite de tiempo de PHP	30
Límite de memoria PHP	256M
Tiempo máximo de entrada	60
Tamaño máximo subida de archivo	2M
PHP post max size	8M
Versión de cURL	7.88.1 OpenSSL/3.0.19
¿Está SUHOSIN instalado?	No
¿Está disponible la biblioteca Imagick?	No
¿Es compatible con enlaces permanentes bonitos?	Sí
Reglas de .htaccess	Se han añadido reglas personalizadas a tu archivo .htaccess.

Base de datos: MySQL

Base de datos

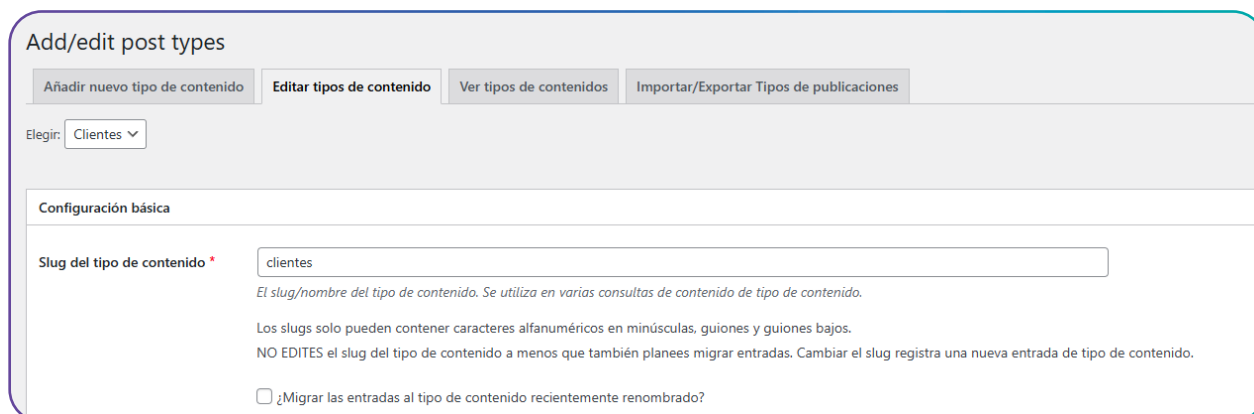
Extensión de la base de datos	mysqli
Versión del servidor	10.11.14-MariaDB-0+deb12u2
Versión del cliente	mysqlnd 8.2.31
Usuario de la base de datos	user
Servidor de la base de datos	localhost
Nombre de la base de datos	wordpress-db
Prefijo de la tabla	wp_

Frontend: HTML, CSS, JavaScript



Plugins para gestión de la base de datos

ACF - Advanced Custom Fields



CPT - Custom Post Types

The screenshot shows the WordPress Custom Post Type (CPT) editor for 'Ficha cliente CDS'. The interface includes a top navigation bar with tabs for 'Grupos de campos', 'Tipos de contenido', 'Taxonomías', and 'Más'. The main area is titled 'Editar grupo de campos' and contains a table of fields.

#	Etiqueta	Nombre	Tipo
1	Código cliente	codigo_cliente	Texto
2	Nombre Empresa *	nombre_empresa	Texto
3	CIF / NIF *	cif	Texto
4	Persona de contacto *	contacto	Texto
5	Correo electrónico *	email_cliente	Correo electrónico
6	Teléfono	telefono_cliente	Texto
7	Estado del cliente	estado_cliente	Selección
8	Servicios contratados	servicios_contratados	Casilla de verificación
9	Nivel de digitalización	nivel_digitalizacion	Selección
10	Técnico responsable *	tecnico_responsable	Selección

La infraestructura permite administración remota segura mediante SSH, facilitando tareas de mantenimiento, monitorización y gestión directa de la base de datos MariaDB alojada en la instancia.

```

Lightsail | eu-west-3 - Google Chrome
eu-west-3.lightsail.aws.amazon.com/ls/remote/eu-central-1/instances/cds-wordpress/terminal?protocol=ssh&region=eu-west-3#

admin@ip-172-26-12-199:~$ cd /var/www
admin@ip-172-26-12-199:/var/www$ mysql -u user -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 22729
Server version: 10.11.14-MariaDB-0+deb12u2 Debian 12

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>

MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| wordpress-db |
+-----+
2 rows in set (0.000 sec)

MariaDB [(none)]>

MariaDB [(none)]> use wordpress-db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [wordpress-db]>
  
```

Acceso a la gestión de la base de datos

```
Database changed
MariaDB [wordpress-db]> show tables;
+-----+
Tables in wordpress-db:
+-----+
wp_actionscheduler_actions
wp_actionscheduler_claims
wp_actionscheduler_groups
wp_actionscheduler_logs
wp_cml2_cookiebanners
wp_cml2_cookies
wp_cml2_disappd
wp_cml2_services
wp_commentmeta
wp_comments
wp_events
wp_ff_scheduled_actions
wp_fluentform_entry_details
wp_fluentform_form_analytics
wp_fluentform_form_meta
wp_fluentform_forms
wp_fluentform_logs
wp_fluentform_submission_meta
wp_fluentform_submissions
wp_links
wp_login_redirects
wp_options
wp_postmeta
wp_posts
wp_snippets
wp_term_relationships
wp_term_taxonomy
wp_termmeta
wp_terms
wp_usermeta
wp_users
wp_wfauditevents
wp_wfblockedplog
wp_wfblockck7
wp_wfconfig
wp_wfcrawlers
wp_wffilechanges
wp_wffilemods
wp_wfhtis
wp_wfhover
wp_wfissues
wp_wfknownfilelist
wp_wflivetraffichuman
wp_wflocs
wp_wflogins
wp_wflogs_2fa_secrets
wp_wflogs_role_counts
wp_wflogs_settings
wp_wfnotifications
wp_wfpendingissues
wp_wfversecache
wp_wfsecurityevents
wp_wfsnppcache
wp_wfstatus
wp_wftrafficates
wp_wfwaifailures
wp_wfforms_logs
wp_wfforms_payment_meta
wp_wfforms_payments
wp_wfforms_tasks_meta
wp_wfmailsmtp_debug_events
wp_wfmailsmtp_tasks_meta
62 rows in set (0.000 sec)
```

```
MariaDB [wordpress-db]> SELECT * FROM wp_users;
+-----+-----+-----+-----+-----+-----+-----+
| ID | user_login | user_pass | user_registered | user_activation_key | user_status | display_name | user_nicename | user_email | user_url |
+-----+-----+-----+-----+-----+-----+-----+
| 1 | cds.admin | 2026-03-14 16:30:37 | 0 | cds.admin | lily.diaz.asir@gmail.com | https://cds
| 2 | cliente_demo | 2026-03-28 22:29:31 | 0 | cliente_demo | cliente@cds-test.es |
| 5 | cliente_daily_dose | 2026-03-30 18:55:22 | 0 | cliente_daily_dose | admin_dd@dailydose.es | https://dai
| 6 | admin_rescate | 2026-04-03 14:16:21 | 0 | Liliana Diaz Lizama | sharpska@hotmail.com |
| 7 | lumen_cliente | 2026-04-08 19:47:01 | 0 | Tecnologias Lumen Norte S.L. | contacto@lumen-norte.es |
| 8 | vega_cliente | 2026-04-08 19:51:46 | 0 | Distribuciones Vega Industrial S.L. | sistemas@vegadistribuciones.es |
| 9 | althema_cliente | 2026-04-08 19:52:35 | 0 | Grupo Althema Servicios Integrales S.L. | administracion@althema-servicios.es |
| 10 | orbital_cliente | 2026-04-08 19:54:12 | 0 | Ingenieria Orbital Delta S.L. | soporte@orbitaldelta.es |
| 11 | eco_cliente | 2026-04-08 19:55:41 | 0 | EcoGestión Urbana Inteligente S.L. | proyectos@ecogestionurbana.es |
| 12 | ruben_soporte | 2026-04-14 19:35:58 | 0 | Rubén Sacristán Martín | rsacristan985@gmail.com |
| 13 | lucia_soporte | 2026-04-14 19:38:25 | 0 | Lucia Campos Flores | luciacampos27@gmail.com |
| 14 | liliana_soporte | 2026-05-09 19:08:45 | 0 | Liliana Diaz | lilysharpska@gmail.com |
| 15 | academia_asir_cliente | 2026-05-15 21:22:19 | 0 | Daniel Delgado | daniel.delgadosoto@educa.madrid.org |
+-----+-----+-----+-----+-----+-----+-----+
13 rows in set (0.000 sec)
```

```
MariaDB [wordpress-db]> DESC wp_users;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| ID | bigint(20) unsigned | NO | PRI | NULL | auto_increment |
| user_login | varchar(60) | NO | MUL | NULL | |
| user_pass | varchar(255) | NO | | | |
| user_nicename | varchar(50) | NO | MUL | NULL | |
| user_email | varchar(100) | NO | MUL | NULL | |
| user_url | varchar(100) | NO | | NULL | |
| user_registered | datetime | NO | | 0000-00-00 00:00:00 | |
| user_activation_key | varchar(255) | NO | | | |
| user_status | int(11) | NO | | 0 | |
| display_name | varchar(250) | NO | | | |
+-----+-----+-----+-----+-----+-----+
```

```
MariaDB [wordpress-db]> DESC wp_usermeta;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| meta_id | bigint(20) unsigned | NO | PRI | NULL | auto_increment |
| user_id | bigint(20) unsigned | NO | MUL | 0 | |
| meta_key | varchar(255) | YES | MUL | NULL | |
| meta_value | longtext | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
```

```
MariaDB [wordpress-db]> DESC wp_fluentform_forms;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| id | int(10) unsigned | NO | PRI | NULL | auto_increment |
| title | varchar(255) | NO | | NULL | |
| status | varchar(45) | YES | | Draft | |
| appearance_settings | text | YES | | NULL | |
| form_fields | longtext | YES | | NULL | |
| has_payment | tinyint(1) | NO | | 0 | |
| type | varchar(45) | YES | | NULL | |
| conditions | text | YES | | NULL | |
| created_by | int(11) | YES | | NULL | |
| created_at | timestamp | YES | | NULL | |
| updated_at | timestamp | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
```

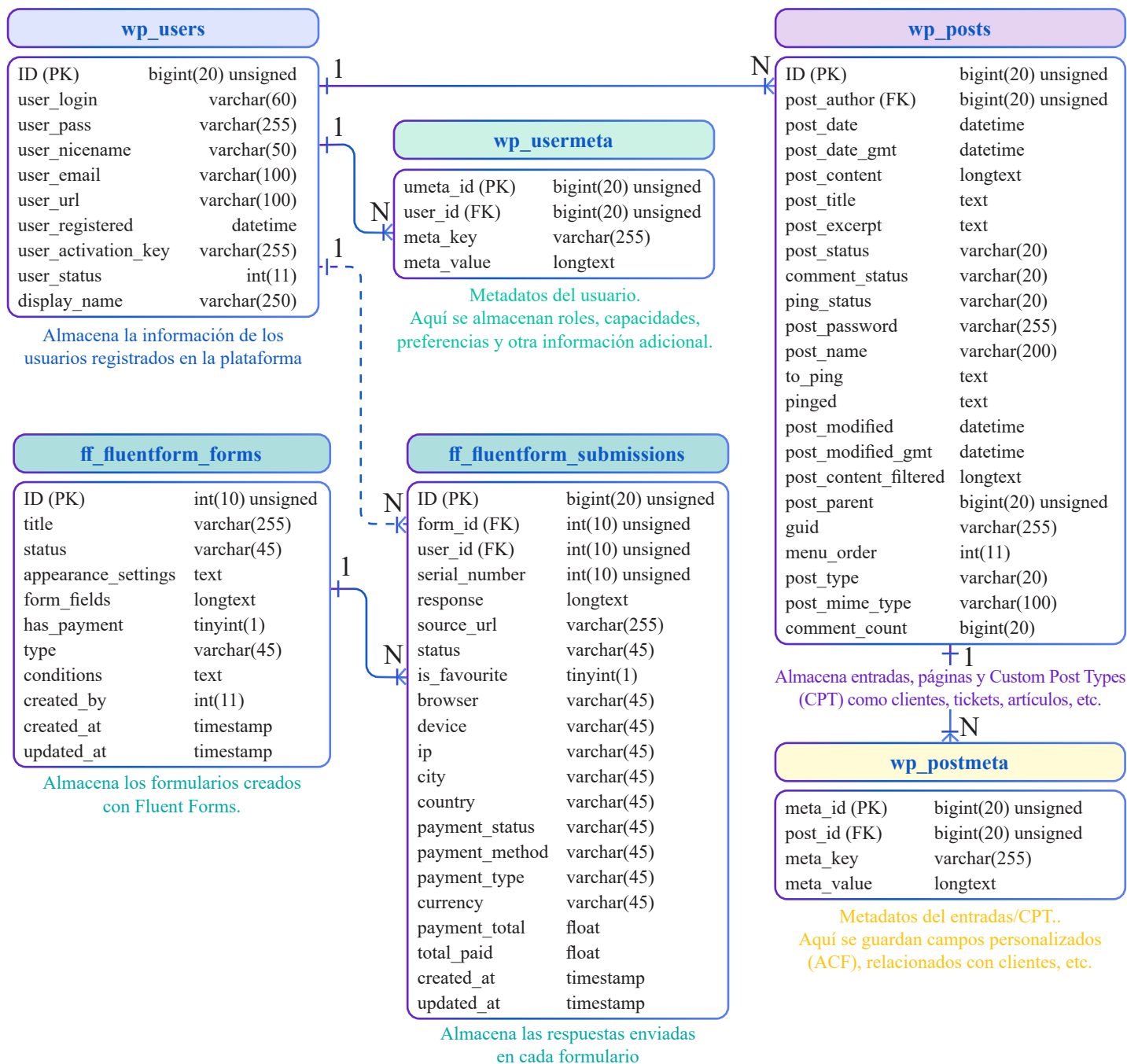
```
MariaDB [wordpress-db]> DESC wp_postmeta;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| meta_id | bigint(20) unsigned | NO | PRI | NULL | auto_increment |
| post_id | bigint(20) unsigned | NO | MUL | 0 | |
| meta_key | varchar(255) | YES | MUL | NULL | |
| meta_value | longtext | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
```

```
MariaDB [wordpress-db]> DESC wp_fluentform_submissions;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| id | bigint(20) unsigned | NO | PRI | NULL | auto_increment |
| form_id | int(10) unsigned | YES | MUL | NULL | |
| serial_number | int(10) unsigned | YES | MUL | NULL | |
| response | longtext | YES | | NULL | |
| source_url | varchar(255) | YES | | NULL | |
| user_id | int(10) unsigned | YES | MUL | NULL | |
| status | varchar(45) | YES | | unread | |
| is_favourite | tinyint(1) | NO | | 0 | |
| browser | varchar(45) | YES | | NULL | |
| device | varchar(45) | YES | | NULL | |
| ip | varchar(45) | YES | | NULL | |
| city | varchar(45) | YES | | NULL | |
| country | varchar(45) | YES | | NULL | |
| payment_status | varchar(45) | YES | | NULL | |
| payment_method | varchar(45) | YES | | NULL | |
| payment_type | varchar(45) | YES | | NULL | |
| currency | varchar(45) | YES | | NULL | |
| payment_total | float | YES | | NULL | |
| total_paid | float | YES | | NULL | |
| created_at | timestamp | YES | | NULL | |
| updated_at | timestamp | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
```

```
MariaDB [wordpress-db]> DESC wp_posts;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| ID | bigint(20) unsigned | NO | PRI | NULL | auto_increment |
| post_author | bigint(20) unsigned | NO | MUL | 0 | |
| post_date | datetime | NO | | 0000-00-00 00:00:00 | |
| post_date_gmt | datetime | NO | | 0000-00-00 00:00:00 | |
| post_content | longtext | NO | | NULL | |
| post_title | text | NO | | NULL | |
| post_excerpt | text | NO | | NULL | |
| post_status | varchar(20) | NO | | publish | |
| comment_status | varchar(20) | NO | | open | |
| ping_status | varchar(20) | NO | | open | |
| post_password | varchar(255) | NO | | | |
| post_name | varchar(200) | NO | MUL | | |
| to_ping | text | NO | | NULL | |
| pinged | text | NO | | NULL | |
| post_modified | datetime | NO | | 0000-00-00 00:00:00 | |
| post_modified_gmt | datetime | NO | | 0000-00-00 00:00:00 | |
| post_content_filtered | longtext | NO | | NULL | |
| post_parent | bigint(20) unsigned | NO | MUL | 0 | |
| guid | varchar(255) | NO | | NULL | |
| menu_order | int(11) | NO | | 0 | |
| post_type | varchar(20) | NO | MUL | post | |
| post_mime_type | varchar(100) | NO | | | |
| comment_count | bigint(20) | NO | | 0 | |
+-----+-----+-----+-----+-----+-----+
```

Esquema relacional - Plataforma CDS (WordPress)

Modelo basado en la estructura real de la base de datos



Leyenda

PK	Clave primaria
FK	Clave foránea
+	Uno
*	Muchos
----	Relación opcional
Relación opcional: un usuario registrado puede enviar múltiples formularios. (ff_fluentform_submissions.user_id referencia a wp_users.ID)	

NOTAS

- WordPress utiliza un modelo EAV (Entidad-Atributo-Valor) a través de las tablas *_meta para almacenar metadatos flexibles.
- Los Custom Post Types (CPT) como clientes, tickets, etc. se almacenan en wp_posts y sus datos adicionales en wp_postmeta.
- Los formularios y sus entradas son gestionados por el plugin Fluent Forms (tablas ff_*).
- Las relaciones mostradas representan la estructura lógica principal utilizada en la plataforma.